

Trajectories of Intrinsic Capacity: Determinants and Associations with Disability

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Abstract

OBJECTIVES: Intrinsic capacity (IC) declines progressively with age, thereby increasing the risk of disability. However, it is less known whether IC trajectories are associated with disability. This study aims to identify the different patterns of IC trajectories in older people, and examine their determinants and associations with Instrumental Activities of Daily Living (IADL).

DESIGN: Cohort study.

SETTING: Community centres in different regions in Hong Kong.

PARTICIPANTS AND MEASUREMENTS: Longitudinal data from community-dwelling older people aged 60 years or above ($n = 1371$) collected between 2016 and 2021 was analysed. Their mean age was 74.5 years, and 78.7% of them were female. Repeated measurements of a set of 14 self-reported items were used to generate IC scores at four time points using a bi-factor model. Latent class growth analysis was performed to identify classes with distinct IC trajectories. The association between class membership and IADL disability was then examined using logistic regression.

RESULTS: Three distinct IC trajectories were identified. The 1st class included those with the highest level of baseline IC and the least declining trajectory, whereas the 3rd class was composed by those with the lowest level of baseline IC and the most declining trajectory. Older age, female gender, lower perceived financial adequacy, living in public or subsidized housing, and chronic diseases were associated with the 3rd class. After adjusting for demographic factors, socioeconomic status, and the number of chronic diseases, the 1st class was more likely to preserve IADL when compared against the 2nd class, with *OR* being 3.179 (95% CI: 2.152–4.793), whereas for the 3rd class, the *OR* was 0.253 (95% CI: 0.178–0.359).

CONCLUSION: Monitoring IC trajectories is of relevance to clinical practice, as it helps shift the focus from treating acute episodes of illness to preserving the functional ability of older people.

Key words: Determinants, disability, functional ability, healthy ageing, intrinsic capacity, trajectories.

Introduction

Globally, life expectancy has increased from 66.8 years in 2000 to 73.4 years in 2019 (1). However, living longer does not entail that the added years are accompanied by healthy lives (2, 3). As age increases, numerous physiological changes occur, which may manifest as decreased physiological reserve and increased risk of disability. In this regard, the World Health Organization (WHO)'s World

Report on Ageing and Health sets the stage for a change in approach to let older people live long and healthy lives (4). The current approach is based on life-course and functional perspectives, since functioning has been suggested to be a better predictor of survival and other health outcomes than individual diseases or comorbidities (5). On the basis of this report, “healthy ageing” is defined as the “the process of developing and maintaining the functional ability that enables well-being in older age”. Central to this healthy ageing model is the concept of intrinsic capacity (IC), which is defined as “the composite of all the physical and mental capacities that an individual can draw on” (4).

IC has been suggested to be an indicator for monitoring and evaluating the progress and impact of the collaborative actions under the United Nations Decade of Healthy Ageing (6). It can be conceptually decomposed into five domains: cognitive, locomotor, sensory, vitality, and psychological (7), and such construct has been validated by various studies in English, Chinese, and Brazilian cohorts (8–11). Further research has demonstrated the ability of IC to predict negative health outcomes (12, 13). To facilitate the identification of those with declining IC for further assessments and early interventions, IC screening has been embedded in the first-step of a novel model of integrated care for older people (ICOPE), which aims at improving, or at least maintaining, older people's IC and functional ability to support healthy ageing, through community-level health and social care interventions (14).

To facilitate the monitoring of health throughout the life-course, a better understanding of IC trajectories is crucial. Nevertheless, the above efforts have primarily focused on the static aspect of IC, and disregarded the fact that individuals may not have identical rates and timing of decline (4). For instance, some older people may maintain high and stable levels of IC, whereas some may experience more declines and fluctuations. To the best of our knowledge, only a few studies have studied IC trajectories and their health impact (15–18). Evidence as to what factors associate with IC trajectories is also sparse. To advance the knowledge on this topic, it is important to probe into the dynamic aspect of IC, which can help improve the understanding of inter-individual heterogeneity in the ageing process, and guide the design of early interventions to steer those with declining or fluctuating IC toward more positive IC trajectories, thereby promoting healthy ageing.

Using a cohort of Chinese older people, the present study aims to identify groups of people sharing similar IC trajectories, to explore the determinants of the IC trajectories, and to examine the association of IC trajectories with Instrumental Activities of Daily Living (IADL) disability over a four-year period of time.

Methods

Design and participants

The current study is part of a larger community primary care project in Hong Kong, which was designed to apply digital technologies to empower older people in health management, as well as to promote integrated care and primary health services that are responsive to the needs of older people. The project services include a longitudinal cohort for identifying the health and social care needs of older people in the community, regular calls and outreach visits, and centre-based activities that can help promote healthy ageing. In this study, data from the longitudinal cohort of Chinese older people were used. The cohort included community-dwelling older people aged 60 years or above recruited from 80 community centres, which were sampled from a comprehensive list of approximately 250 community centres in Hong Kong (19). Of the 80 centres, 19 (24%) were selected within Hong Kong Island, 26 (33%) within Kowloon, and 35 (44%) within the New Territories. The sampling distribution was set to match with the proportions of older people aged 60 years or above residing in the three regions (i.e., 20%, 37%, and 43%), respectively, according to the results of the 2011 Population Census (20).

The longitudinal survey was carried out over four waves, namely 2016/2017 (wave 1), 2017/2018 (wave 2), 2018/2019 (wave 3), and 2019/2020 (wave 4). The initial cohort recruited at wave 1 consisted of 2498 older people. They were given a tablet computer to complete the survey, which covered items for capturing IC, demographic factors, socioeconomic status, health conditions, medications, incontinence, self-rated health, IADL, and use of health services, and were invited to retake the survey annually. The flow of participants is shown in Supplementary Figure S1. Of the 2498 participants, 1601 (64.1%) had also participated in the one-, two-, and three-year follow-ups. Participants were excluded from the present study if they had incomplete data on the exposure measures (IC), outcome (IADL), or any of the covariates ($n = 230$). Ultimately, a total of 1371 participants were included in this study. Compared with participants included in the analysis, excluded participants whose demographic information was available ($n = 912$) were older, had a lower level of education, were less likely to be married, were more likely to be living alone, recipients of financial assistance, and disabled (Supplementary Table S1). All participants have given informed consent.

Intrinsic capacity

IC was assessed at baseline and at every follow-up wave. A set of 14 self-reported items were measured, which covered

the five domains of IC: (i) cognitive capacity was assessed with the 5-item Abbreviated Memory Inventory for Chinese (AMIC) (21). The items are yes-no questions including, for instance, “do you always forget where things are placed?”, “do you forget your acquaintances’ names when you meet them?”, and “do you feel that your memory is worse than your peers?”. The total AMIC score ranges from 0 to 5, with 1 point given to each item answered “yes”; (ii) locomotor capacity was assessed with the yes-no questions “do you have any difficulty walking up 10 steps alone without resting and without aids?” and “do you have any difficulty walking several hundred yards alone and without aids?”, which are items extracted from the 5-item FRAIL scale (22); (iii) vitality was assessed by the yes-no question “do you have any weight loss of $\geq 5\%$ within the past 6 months?” and “how much difficulty do you have in lifting and carrying 10 pounds?” (0 = none; 1 = some; 2 = a lot or unable); which is an item extracted from the SARC-F scale (23); (iv) sensory (vision) capacity was assessed using an item similar to that described in the ICOPE Handbook, i.e., “do you see things clearly?” (1 = very poor; 6 = very good), whereas sensory (hearing) capacity was assessed using the question “do you hear things clearly?” (1 = very poor; 6 = very good); (v) psychological capacity was assessed with three questions including “are you satisfied with your life?” (yes/no), “are you happy with your life?” (0 = the most unhappy; 8 = the happiest), and “do you find purpose or meaning in your life?” (yes/no), for capturing the evaluative, hedonic, and eudemonic well-being of the participants, respectively (24).

Disability

Disability was assessed at baseline and at every follow-up wave, using five items (telephoning, use of transportation, shopping, food preparation, and financial behavior) extracted from the Lawton’s IADL scale (25) with a modified anchor and scoring system. For each item, participants were asked if they were able to complete a certain daily task independently (score = 1) or with occasional help (score = 2), or whether they were unable to perform the task in the past three months (score = 3). Those who scored a 2 or 3 in any of the items were classified as having IADL disability. It should be noted that IADL was assessed instead of Basic Activities of Daily Living (BADL), as the former includes more complex and higher-order activities that are related to independent living in the community, which may be more sensitive to early decline in IC, particularly in the cognitive domain.

Demographic factors

Four demographic factors were assessed: (i) age, (ii) gender, (iii) marital status, coded as married or not married, and (iv) living arrangement, coded as living alone or living with others.

Socioeconomic status

Socioeconomic status was assessed with four different self-reported indicators: (i) educational attainment, coded as no

formal schooling, primary, secondary, or tertiary, (ii) perceived financial adequacy, coded as very adequate, adequate, just right, inadequate, or very inadequate, (iii) financial assistance status, coded as recipient or non-recipient of financial assistance of social security, and (iv) type of housing, coded as public, subsidized, or private housing. Type of housing was used as a proxy of wealth, given its higher stability and stronger influence on living standards over the long term, while education and current income ignore the aspect of accumulated resources (26). In this study, it was assumed that older people living in public housing were the least wealthy, and those living in private housing the most wealthy, as rents and prices of public or subsidized housing are generally lower than those of private housing. It is also the Hong Kong Government's housing policy to provide public housing for lower-income residents and subsidized public rental housing for those who cannot afford private housing. Previous studies showing associations between housing related variates (e.g., relative residential property value) and health support this assumption (27).

Health conditions

The number of chronic diseases diagnosed by a western medical practitioner was obtained. The types of chronic diseases being considered included hypertension, diabetes, high cholesterol, heart diseases, stroke, chronic obstructive pulmonary disease, and renal diseases.

Statistical analysis

The lavaan package (version 0.6-11) in R (version 4.2.0) was used to fit a bi-factor model on the 14 self-reported items, which were reverse scored, as appropriate, in advance. To begin, a multi-group confirmatory factor analysis was conducted, by treating the time index as the grouping variable, to check whether the same factor structure was applicable to all time points. It is sufficient to consider the configural, metric, and scalar invariance (i.e., same factor structure, loadings, and intercepts), since it already enables the comparison of IC scores across different time points. Results of this multi-group confirmatory factor analysis, which indeed ascertained the longitudinal measurement invariance, are shown in Supplementary Table S2. Based on this, the set of estimates obtained from the baseline data was used to generate the IC scores in all time points. To provide a preliminary check on construct validity, a linear mixed-effects model was used to assess whether the IC scores were sensitive to ageing. Subsequently, Mplus (version 8.8) was used to perform a latent class growth analysis on the IC scores, in which the variances and covariance of the growth factors were both fixed to zero. To select the number of latent classes, a number of widely used measures were referred to, including the Lo-Mendell-Rubin adjusted likelihood ratio test, the sample-size adjusted Bayesian information criteria, entropy, and average posterior probabilities for the latent class membership (required to be $> .7$). Ultimately, a three-class solution was deemed to be the optimal one. Based on the estimated class probabilities, a latent class membership

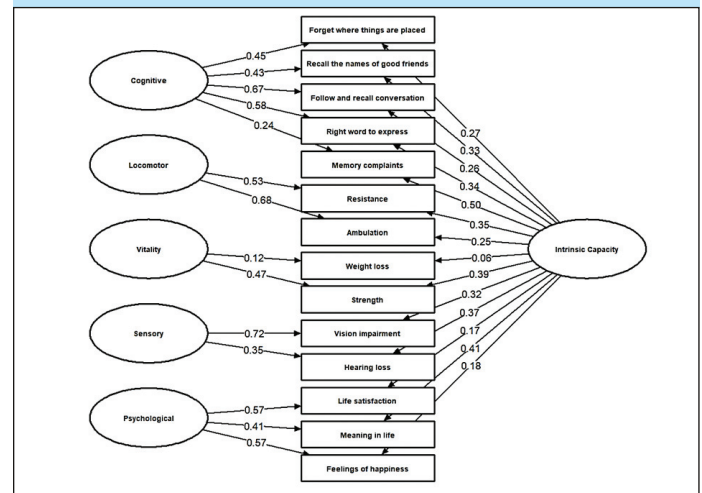
variable was then created. This variable was first treated as the outcome variable in a proportional odds ordinal logistic regression model to examine its determinants. Finally, the variable was treated as an independent variable in a logistic regression model to probe into its association with IADL, after adjusting for demographic factors (age, gender, marital status, and living arrangement), socioeconomic status (educational attainment, perceived financial adequacy, financial assistance status, and type of housing), and number of chronic diseases.

Results

Validity of IC scores

A total of 1371 older people participated in all four waves of the study. A bi-factor model was used to generate IC scores at the four time points, which achieved a good fit on the data, with comparative fit index (CFI) = .987, Tucker-Lewis index (TLI) = .977, root mean square error of approximation (RMSEA) = .023 (90% CI: .014–.030), and standardized root mean square residual (SRMR) = .019 (Figure 1). Age ($\beta = -0.023$, 95% CI: -0.030 – -0.016), female gender ($\beta = -0.171$, 95% CI: -0.283 – 0.059), living in public ($\beta = -0.220$, 95% CI: -0.336 – -0.104) or subsidized housing ($\beta = -0.246$, 95% CI: -0.386 – -0.106), and number of chronic diseases ($\beta = -0.059$, 95% CI: -0.096 – 0.021) were associated with lower IC scores, whereas perceived financial adequacy ($\beta = 0.312$, 95% CI: 0.250 – 0.375) was associated with higher IC scores (Supplementary Table S3). These findings support the construct validity of the IC scores.

Figure 1. Bi-factor model



Note: CFI = .987, TLI = .977, RMSEA = .023 (90% CI: .014–.030), SRMR = .019. The 14 manifest variables in the above model were measured using self-reported items.

Classes of IC trajectory

Three distinct IC trajectories were identified (Supplementary Figure S2). The 1st class included those with the highest level of baseline IC and the least declining trajectory. The 2nd class's trajectory ran largely in parallel with that of the 1st class, but started with a lower level of baseline IC. The 3rd class was composed by those with the lowest level of baseline IC and the

Table 1. Demographic profiles of the latent classes

Variable	Mean (SD) / Count (%)			p
	Class 1 (n = 497)	Class 2 (n = 643)	Class 3 (n = 231)	
Demographic factors				
Age, years	72.88 (6.66)	74.78 (6.85)	77.39 (6.80)	< .001
Gender				
Male	125 (25.2)	131 (20.4)	36 (15.6)	
Female	372 (74.8)	512 (79.6)	195 (84.4)	
Marital status				< .001
Married	274 (55.1)	321 (49.9)	74 (32.0)	
Not married	223 (44.9)	322 (50.1)	157 (68.0)	
Living arrangement				< .001
Alone	139 (28.0)	211 (32.8)	117 (50.6)	
With others	358 (72.0)	432 (67.2)	114 (49.4)	
Socioeconomic status				
Educational attainment				< .001
No formal schooling	97 (19.5)	170 (26.4)	85 (36.8)	
Primary	226 (45.5)	315 (49.0)	94 (40.7)	
Secondary/Tertiary	174 (35.0)	158 (24.6)	52 (22.5)	
Perceived financial adequacy	3.14 (0.75)	2.89 (0.63)	2.71 (0.60)	< .001
Financial assistance status				< .001
Recipient	74 (14.9)	128 (19.9)	61 (26.4)	
Non-recipient	423 (85.1)	515 (80.1)	170 (73.6)	
Type of housing				< .001
Public housing	280 (56.3)	419 (65.2)	166 (71.9)	
Subsidized housing	89 (17.9)	114 (17.7)	41 (17.7)	
Private housing	128 (25.8)	110 (17.1)	24 (10.4)	
Health conditions				
Number of chronic diseases	1.25 (1.06)	1.56 (1.16)	1.53 (1.11)	< .001

Note: *p*-values about group differences were obtained using regression *F*-test and Pearson's χ^2 test as appropriate.

most declining trajectory. The profile of participants at baseline for each class is shown in Table 1. Participants in the 1st class were younger, more likely to be male, married, living with others, had higher socioeconomic status (in terms of educational attainment, perceived financial adequacy, financial assistance status, and type of housing), and less chronic diseases (all *ps* < .01).

Determinants of IC trajectory class

In an ordinal logistic regression analysis, age (*OR* = 0.945, 95% CI: 0.929–0.961), female gender (*OR* = 0.633, 95% CI: 0.479–0.834), living in public housing (*OR* = 0.578, 95% CI: 0.433–0.770) or subsidized housing (*OR* = 0.562, 95% CI: 0.397–0.794), and number of chronic diseases (*OR* = 0.862, 95% CI: 0.787–0.945) were associated with a lower level of IC trajectory class (i.e., 3rd class), while perceived financial adequacy (*OR* = 1.929, 95% CI: 1.645–2.267) was associated with a higher level of IC trajectory class (i.e., 1st class) (Table 2).

IC trajectory class and IADL disability

Significant association between IC classes and IADL disability at the three-year follow-up was found. Using the

2nd class as the reference class, which comprised the highest percentage of participants (46.9%), and after adjusting for demographic factors, socioeconomic status, and number of chronic diseases, the 1st class was more likely to preserve IADL, with *OR* being 3.179 (95% CI: 2.152–4.793), whereas for the 3rd class, the *OR* was 0.253 (95% CI: 0.178–0.359) (Table 3). The 1st class also had the highest probability of maintaining intact IADL over time (Figure 2). Notably, the 1st and the 2nd classes tended to have similar trajectories despite a different level of IC at the baseline, whereas the gap existing between the 3rd class and the other two classes at the baseline tended to increase over time.

Discussion

The present study identified three distinct latent IC trajectories across three years of follow-ups. Participants in the 1st class exhibiting the highest level of baseline IC and the least declining trajectory were associated with a higher probability of maintaining intact IADL. Older age, female gender, lower perceived financial adequacy, living in public or subsidized housing, and higher number of chronic diseases were associated to lower probability of belonging to the 1st class. These results suggest that maintaining IC at a high level could help prevent disability in later life.

Table 2. Proportional odds ordinal logistic regression on latent class membership ($n = 1371$)

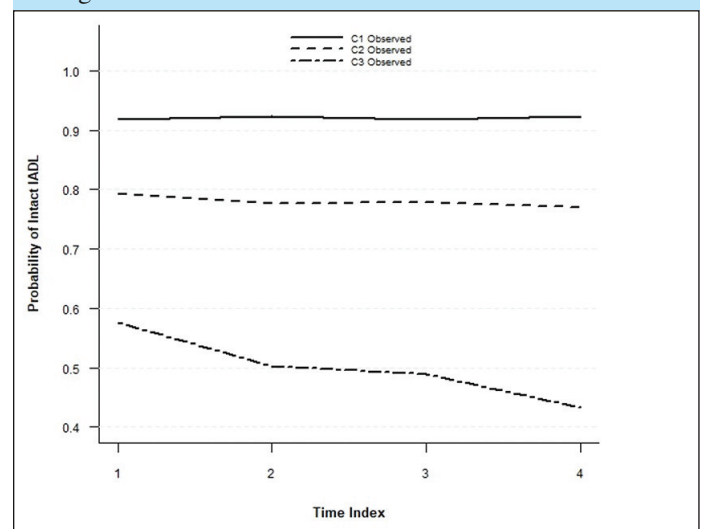
Variable	OR	95% CI
<i>Demographic factors</i>		
Age, years	0.945	(0.929, 0.961)
Gender		
Male	Ref. category	
Female	0.633	(0.479, 0.834)
Marital status		
Married	Ref. category	
Not married	0.822	(0.638, 1.060)
Living arrangement		
Alone	0.820	(0.624, 1.076)
With others	Ref. category	
<i>Socioeconomic status</i>		
Educational attainment		
No formal schooling	0.924	(0.673, 1.270)
Primary	1.046	(0.808, 1.354)
Secondary/Tertiary	Ref. category	
Perceived financial adequacy	1.929	(1.645, 2.267)
Financial assistance status		
Recipient	0.941	(0.705, 1.258)
Non-recipient	Ref. category	
Type of housing		
Public housing	0.578	(0.433, 0.770)
Subsidized housing	0.562	(0.397, 0.794)
Private housing	Ref. category	
<i>Health conditions</i>		
Number of chronic diseases	0.862	(0.787, 0.945)

Note: Class 3 was treated as the lowest level, whereas class 1 was treated as the highest. An OR of value greater than 1 indicates higher probability of belonging to class 1, vice versa.

Determinants of IC trajectories

The likelihood of belonging to the 1st class was lower among older participants. Indeed, among the demographic variables, age is a definite and non-modifiable risk factor for IC declines. This can be explained by, for instance, the well-established associations between advanced age and deteriorations in the cognitive (28), locomotor (29), and sensory (30) domains of IC. On the other hand, decline in IC may also depend on socioeconomic status and health conditions. After adjusting for age and other demographic factors, our results showed that lower perceived financial adequacy and wealth (using type of housing as a proxy measure) were associated with lower probability of being in the 1st class of IC trajectories. These findings aligned with the studies that have provided evidence regarding the socioeconomic determinants of various health outcomes in older adults, including health trajectories (31), frailty (32), and onset of disability (33). Two themes

are central to the explanation of the associations of financial adequacy and wealth with health. First, older people with higher perceived financial adequacy and wealth have a better economic status, which enables a more effective use of money to secure goods and services (e.g., healthcare) needed for a healthy life (34). Second, those with lower perceived financial adequacy and wealth may perceive having a lower social status, which may lead to chronic mental distress and somatic disease (35). In contrast to common belief, however, our results showed that there is a limited impact of educational attainment on IC trajectories. Perhaps the health benefits of education are mostly explained away by perceived financial adequacy and wealth. Further analyses of our data found that the relationship between educational attainment and IC trajectories was indeed fully mediated by perceived financial adequacy and type of housing (result not shown). With regard to medical factors, our findings revealed that older people with more chronic diseases were less likely to experience better IC trajectories. This finding is consistent with previous results, which revealed a negative association between multi-morbidity and health trajectories (31, 36). Nevertheless, the association between chronic diseases and IC could be reciprocal. A previous study found that decline in IC, or more specifically, a constellation of three or more impaired IC domains, could predict chronic conditions (e.g., incontinence) and polypharmacy in older people (37). Consistent with the concept of frailty (38), losses of IC in specific domains (e.g., vitality and locomotor) can be linked to specific pathophysiologies (e.g., dysregulation of innate immune system), which may in turn increase the risk of developing chronic diseases.

Figure 2. Trajectories in Instrumental Activities of Daily Living

Note: C1: Class 1; C2: Class 2; C3: Class 3.

Table 3. Logistic regression for the association between the latent classes and follow-up Instrumental Activities of Daily Living ($n = 1371$)

Variable	OR	95% CI
Latent class		
Class 1	3.179	(2.152, 4.793)
Class 2	Ref. category	
Class 3	0.253	(0.178, 0.359)
Baseline IADL		
Disabled	0.310	(0.225, 0.426)
Intact	Ref. category	
<i>Demographic factors</i>		
Age, years	0.950	(0.928, 0.972)
Gender		
Male	Ref. category	
Female	0.752	(0.504, 1.109)
Marital status		
Married	Ref. category	
Not married	1.526	(1.065, 2.196)
Living arrangement		
Alone	1.123	(0.765, 1.649)
With others	Ref. category	
<i>Socioeconomic status</i>		
Educational attainment		
No formal schooling	1.186	(0.766, 1.837)
Primary	1.042	(0.722, 1.500)
Secondary/Tertiary	Ref. category	
Perceived financial adequacy	0.821	(0.653, 1.030)
Financial assistance status		
Recipient	1.033	(0.697, 1.542)
Non-recipient	Ref. category	
Type of housing		
Public housing	1.114	(0.742, 1.660)
Subsidized housing	1.122	(0.692, 1.821)
Private housing	Ref. category	
<i>Health conditions</i>		
Number of chronic diseases	1.021	(0.899, 1.162)

Note: IADL: Instrumental Activities of Daily Living. Number of participants with intact follow-up IADL = 1054 (76.9%). Nagelkerke's pseudo $R^2 = .291$. Likelihood ratio test for the latent class variable was significant, $\chi^2(2) = 127.31$, $p < .001$. An OR of value greater than 1 indicates higher probability of having intact IADL at follow-up, vice versa.

Classes of IC trajectory and IADL disability

Previous studies have shown that declines in IC was associated with higher risk of disability or dependency (8-10, 12, 13). However, most of these studies relied on a single assessment of IC. Some of the only exceptions include: (i) the Yale PEP study which, based on repeated measurements of IC over 20 years, demonstrated that IC declined progressively, and that the most recent IC value could predict future negative health outcomes, including onset of IADL disability, long-term nursing home stay, and mortality (16); (ii) a study in Hong Kong revealing that declines in IC was associated with a faster decline in IADL over time (15). However, these studies did not examine how different trajectories of IC might affect

the prediction of disability. Building on the previous work, the present study found three stably declining trajectories starting at different initial IC levels, and that the 1st class was associated with a higher probability of maintaining intact IADL, after adjusting for demographics, socioeconomic status, and number of chronic diseases. This finding is consistent with the results from the WHO study on global AGEing and adult health (SAGE) in Mexico and another study in Taiwan that the class with better IC trajectories exhibited lower disability scores (17, 18). Furthermore, the present study showed that the trends in disability diverged, i.e., the gap between the 2nd and the 3rd classes widened across the follow-up years. A plausible explanation for this phenomenon is that people with better IC trajectories may be characterized by better physiological reserves, which allow them to recover more effectively from age- and disease-related changes, and thereby maintain independence. In contrast, when loss of IC exceeds a certain threshold or occurs in two or more domains, a vicious cycle may be formed, leading to a progressive decline in IC. Further analyses of our data found that most of those in the 3rd class had impaired locomotor and vitality domains (Supplementary Figure S3). Furthermore, impairments in these two domains tended to develop first (Supplementary Table S4), and were strongly predictive of disability (Supplementary Table S5). Nevertheless, locomotor and vitality impairments rarely developed alone, but rather co-occurred with other manifestations (e.g., visual impairment, hearing loss, lower psychological well-being) (Supplementary Figure S3). Using a Bayesian network approach, further exploratory analysis on the inter-relationships between the five domains found that the vitality domain was the ancestor (if not the parent) of all other IC domains, whereas locomotor, sensory, and cognitive domains could act as intermediate factors in the pathway between vitality and psychological capacity (Supplementary Figure S4). Taken together, the finding raises the possibility that decline in vitality may presage declines in the capacity of other IC domains, and is hence involved in the threshold transition toward IADL disability (39, 40). Thus, strategies to improve functional ability should consider the dynamic functioning across multiple domains of IC.

Strengths and limitations

The current study is, to our best knowledge, one of the only few studies to investigate how different patterns of IC trajectories are associated with disability in a dynamic manner. Regarding the strengths of the present study, following the WHO conceptual framework and some recent research (8-11), a bi-factor model was used to capture the latent construct of IC, which did not only result in an extremely satisfactory fit, but was also significantly better than a traditional second-order model (result not shown). This, in conjunction with the scalar longitudinal invariance, showed that the bi-factor model can indeed provide a robust measurement of IC and allow it to be compared over time. In addition, latent class growth analysis was used to investigate the association between IC trajectories and disability. In contrast, previous studies primarily

aimed at either examining the determinants of IC trajectories or predicting health outcomes using the IC score measured at a single time point. However, this study also had several limitations. First, participants who dropped out from the study or with incomplete data were excluded, which may have led to selection bias as the excluded individuals might be relatively less robust or even be those who died during the study. Second, the follow-up time of the present study is short, which may not allow the long-term intra-individual variability in IC to be captured, especially since fluctuations in health and disablement process can be higher in the later life (41). Third, since the measures used to determine IC were self-report in nature, recall bias may be present and such a systematic source of error may not be appropriately taken into account by the measurement model. Fourth, the effects of time-varying covariates on IC trajectories were not examined because data were not available in all the follow-ups. Last, data on lifestyle and health behaviors (e.g., physical activities, smoking, alcohol drinking, and body mass index) and environmental factors, which were found to be associated with IC or health trajectories (15, 42), and information regarding acute stressors that may have contributed to functional deterioration, were not available.

Conclusions

This study provides evidence regarding the inter-individual heterogeneity in the ageing process and knowledge on the determinants of IC trajectories. Our findings also suggest that maintaining IC at a higher level over time can reduce the risk of disability. This implies that monitoring IC indeed has its clinical utility. As life expectancy increases, healthcare systems should not merely focus on treating acute episodes of illness, but also maintaining IC over time, as well as preserving the functional ability for those with declining IC.

Ethical standards: The study was approved by the Survey and Behavioural Research Ethics Committee of The Chinese University of Hong Kong.

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Conflicts of Interest: None declared.

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